

FREIBURG, Germany

WMO: 108030

Lat: 48.0233N

Lon: 7.8342E

Elev: 238

StdP: 98.50

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-6.9	-13.7	1.2	-8.2	-11.0	1.5	-5.4	13.2	8.2	12.2	9.0	1.7	350	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	31.1	20.1	29.2	19.3	27.4	18.6	20.9	28.7	20.1	27.5	19.4	26.1	3.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.6	24.1	17.5	12.9	23.2	16.7	12.2	22.6	61.3	28.9	58.6	27.6	56.0	26.2	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.5	8.3	DB	-10.1	34.8	4.1	1.4	-13.0	35.8	-15.5	36.6	-17.8	37.4	-20.8	38.4	(4)
(5)			WB	-10.6	22.9	4.0	0.7	-13.5	23.5	-15.8	23.9	-18.1	24.3	-21.0	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	2.1	2.6	7.0	10.1	14.7	18.2	20.9	20.6	16.9	11.9	6.2	3.9
	DBStd	7.81	5.53	5.18	3.94	3.69	3.88	3.83	3.33	3.33	3.34	3.68	4.32	4.65
	HDD10.0	958	247	211	107	43	7	0	0	0	0	22	126	195
	HDD18.3	2819	502	441	351	246	121	50	13	17	67	203	364	446
	CDD10.0	1442	3	3	15	47	154	247	338	328	207	80	12	7
	CDD18.3	262	0	0	0	1	9	46	93	87	23	2	0	0
	CDH23.3	1918	0	0	1	6	75	334	740	633	119	11	0	0
	CDH26.7	548	0	0	0	0	5	85	242	196	19	2	0	0
Wind	WSAvg	3.1	3.5	3.2	3.7	3.3	3.1	3.2	2.9	2.9	2.9	2.9	2.9	3.0
Precipitation	PrecAvg	962	63	53	63	78	109	123	95	104	70	65	74	62
	PrecMax	1222	101	122	127	181	244	219	162	289	166	172	144	168
	PrecMin	682	12	19	20	30	40	25	35	30	17	13	24	2
	PrecStd	136	25	26	28	40	48	47	38	57	42	42	31	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	17.0	22.2	24.3	27.3	31.5	34.0	33.9	29.0	24.4	17.7	16.9
		MCWB	9.6	10.3	12.4	13.7	16.3	20.5	21.2	20.5	18.1	16.2	12.0	10.3
	2%	DB	12.5	13.9	17.1	21.1	25.5	29.1	31.2	31.1	26.4	20.9	15.5	14.6
		MCWB	8.5	9.0	10.2	11.8	15.6	19.4	20.3	19.6	18.0	14.5	11.5	9.9
	5%	DB	10.8	11.3	14.6	18.8	23.6	27.0	29.4	28.7	24.4	18.6	13.6	12.5
		MCWB	7.7	7.9	9.1	11.2	15.0	18.4	19.7	19.0	17.3	13.8	10.6	9.3
	10%	DB	9.3	9.5	12.9	16.6	21.6	25.1	27.5	26.8	22.7	16.7	12.1	10.6
		MCWB	6.7	6.9	8.4	10.0	14.4	17.5	19.2	18.4	16.6	13.1	9.6	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	10.7	12.7	14.1	18.1	21.5	22.4	21.5	19.9	16.8	13.1	12.3
		MCDB	14.2	16.0	20.4	21.8	24.5	28.9	30.0	30.2	26.0	22.2	16.2	14.3
	2%	WB	9.0	9.5	11.0	12.9	16.9	20.2	21.3	20.5	18.8	15.7	12.0	10.5
		MCDB	11.6	13.1	16.2	19.8	22.7	27.4	29.1	28.5	24.4	19.9	14.5	13.5
	5%	WB	8.1	8.2	9.8	11.9	15.7	19.2	20.4	19.7	17.9	14.4	11.0	9.3
		MCDB	10.6	10.8	13.5	16.8	21.4	25.7	27.9	27.0	23.1	17.7	13.2	12.1
	10%	WB	6.7	7.1	8.8	10.9	14.9	18.1	19.6	19.0	17.1	13.4	9.6	7.9
		MCDB	9.1	9.6	12.2	15.8	20.4	24.1	26.4	25.4	21.8	16.1	12.0	10.8
Mean Daily Temperature Range	5% DB	MDBR	4.9	6.0	7.2	8.3	9.0	8.8	9.6	9.7	8.7	7.0	5.6	4.7
		MCDBR	6.4	8.8	10.1	12.2	12.2	11.8	12.7	12.7	11.3	9.8	8.3	7.2
		MCWBR	4.6	5.6	5.5	6.0	5.9	5.4	5.7	5.3	5.1	5.2	5.3	5.1
	5% WB	MCDBR	6.0	7.8	8.4	10.6	10.4	11.1	11.4	11.4	10.2	8.7	7.3	6.8
		MCWBR	4.7	5.4	5.1	5.6	5.6	5.3	5.4	5.1	5.1	4.9	5.3	5.2
Clear-Sky Solar Irradiance	taub	0.313	0.338	0.372	0.408	0.418	0.424	0.422	0.417	0.392	0.369	0.335	0.310	
	taud	2.440	2.356	2.304	2.230	2.248	2.271	2.270	2.301	2.347	2.399	2.446	2.441	
	Ebn at Noon	761	814	839	841	846	842	837	824	809	768	725	715	
	Edh at Noon	66	88	109	129	132	130	128	119	104	84	66	59	
All-Sky Solar Radiation	RadAvg	1.01	1.80	3.02	4.30	5.04	5.84	5.52	4.80	3.63	2.22	1.16	0.82	
	RadStd	0.14	0.29	0.45	0.62	0.53	0.51	0.51	0.48	0.36	0.20	0.14	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (98 neighbors)	N/A	N/A	N/A	N/A	+0.39	+0.31	N/A	N/A	+49	+17

Nomenclature: See separate page